Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 1 Random events and its probability

Section 1 random events

- Introduction to probability
- Random phenomena
- Random experiment
- Sample space and sample point
- Events

Introduction

- The use of probability to measure uncertainty and variability dates back hundreds of years.

- It is generally believed that the mathematical theory of probability was started by the French mathematicians Blaise Pascal(1623 - 1662)and Pierre Fermat(1601 -1665) when they succeeded in deriving exact probabilities for certain gambling problems involving dice.

- Probability theory is an important tool in most areas of engineering, science, and management, such as weather broadcasting, earthquake broadcasting.

Determinant and random phenomena

- **Deterministic phenomena** are those inevitable phenomena under certain condition(s). Conditions determine results!

  Examples of deterministic phenomena: daily sunrises and sunsets, tides at sea shores, phases of the moon, seasonal changes in weather, annual flooding of the Yellow River and etc.

- **Random phenomena** are those phenomena whose outcome may vary with some potential law. Conditions do not completely determine results!

  Examples of random phenomena: results of coin tosses, results of rolling dice, results of horse races, prediction of the highest and lowest temperature tomorrow and etc.

Determinant and random phenomena

- **Random event** **happens occasionally in a single test, but in massive repeated tests, the outcomes has a certain statistical regularity.** The theory of probability is a mathematical discipline that studies this essential law of random phenomenon.
- How to study random phenomena ? ---random experiment! What is random experiments?

Examples of experiment

- E1: Tossing a fair coin, observing conditions of frontage H and the reverse side T.
- E2: Tossing a fair coin three times, observing conditions of frontage H and the reverse side T. (Tossing three distinct coins once.)
- E3: Tossing a fair coin three times, observing times of H.
- E4: Rolling a fair dice, observing the number that shows on the top.
- E5: Recording the number of traffic accidents which occurred in a specified location of Beijing within a certain period of time.
- E6: Recording the lifetime of an electronic device, or of an electrical appliance, etc.
- E7: Recording the highest and the lowest score of a course.

Random experiment

- The experiments can be repeated under the same condition(s)
- Possible outcomes are not unique,
- but we can know all the possible outcomes in advance.
- We don't know what results will appear before the experiment.

Definition 1.1.1: The experiment which satisfies the above conditions is a random experiment.

Note：

- random experiment is a widely used term, including all kinds of scientific experiments as well as investigation, observation and measurement of certain target.

Random experiment

Example1.1.1：

- Tossing a fair coin, observing conditions of frontage H and the reverse side T. Is this a random experiment?

- The experiments can be repeated under the same condition(s)? √
- Possible outcomes are not unique? √
- We can know all the possible outcomes in advance. √
- We don’t know what results will appear before the experiment. √

Which are random experiments?

- E1: Tossing a fair coin, observing conditions of frontage H and the reverse side T.
- E2: Tossing a fair coin three times, observing conditions of frontage H and the reverse side T. (Tossing three distinct coins once.)
- E3: Tossing a fair coin three times, observing times of H.
- E4: Rolling a fair die, observing the number that shows on the top.
- E5: Recording the number of traffic accidents which occurred in a specified location of Beijing within a certain period of time.
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- E7: Recording the highest and the lowest score of a course.

Sample Space and sample point

Definition 1.1.2 **Sample space:** The sample space of an experiment is a set of all possible outcomes (points). The sample space is denoted by S or Ω.
The elements in the sample space are called sample points.

Note: The sample space is deterministic event.

We define an empty set with no element as an impossible event, indicated by Φ

Events

Definition 1.1.3 **Events:** An event is a collection or any subset of outcomes/elements from the sample space Ω. Event that contains only one sample point is called a elementary event.

Example: Rolling a fair die, observing the number that shows on the top.

Events

Example: Rolling a fair die, observing the number that shows on the top.

In this random experiment, ‘the number is 1’, ‘the number is 2’, ‘the number is odd’, ‘the number is even’, ‘the number is less than 4’ are **random events** or **events**.

Write down the sample points, sample space, elementary event, Event A- the number is odd, event B –the number is even.

Sample point: ω_i corresponding to number i on the top, $i = 1,2,3,4,5,6$

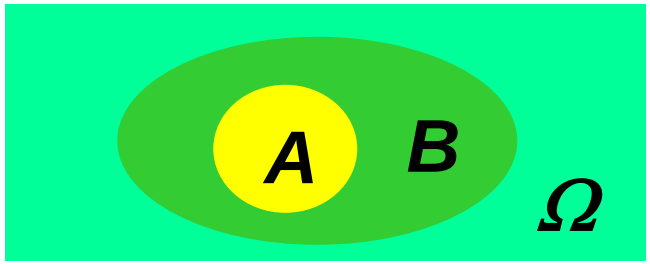
Sample space: $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\}$

$A = \{\omega_1, \omega_3, \omega_5\}$ $\qquad$ $B = \{\omega_2, \omega_4, \omega_6\}$

Section 2 Operations and relations of events

Assuming that the sample space of a random trial E is Ω, and $A, B, A_k (k = 1,2,4 \dots)$ are subsets of Ω.

(**Subevent**) If A ⊂ B, we call the event B contains the event A or the event A is the subevent of B. This means that the occurring of event A inevitably leads to the occurring of the event B.

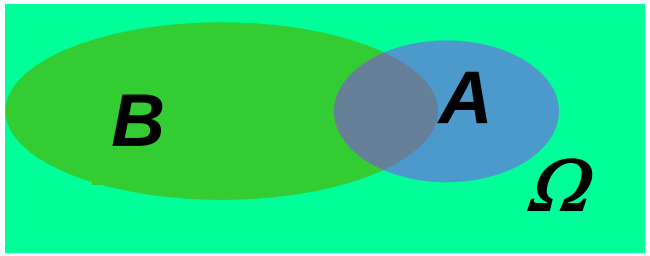


All boys' homework is finished → a boy's homework is finished.
a boy's homework is finished → all boys' home work is finished. X

Section 2 Operations and relations of events

Assuming that the sample space of a random trial E is Ω, and $A, B, A_k (k = 1,2,4 \ldots)$ are subsets of Ω.

(Union) The union of two events A and B, denoted by the symbol $A \cup B = \{x | x \in A \ or \ x \in B\}$, is the event containing all elements that belong to A or B or both.



All students' homework is the union of all boys' homework is finished and all girls' homework is finished.

Note: $\bigcup_{k=1}^{n} A_k$ is the union of events $A_k (k = 1,2,3 \ldots)$.

Operation property of union and intersection

$$\boldsymbol{A} \cup \boldsymbol{A} = \boldsymbol{A}, \quad A \cup \Omega = \Omega, \quad \boldsymbol{A} \cup \varnothing = \boldsymbol{A},$$

$$\boldsymbol{A} \cap \boldsymbol{A} = \boldsymbol{A}, \quad A \cap \Omega = A, \quad \boldsymbol{A} \cap \varnothing = \varnothing.$$

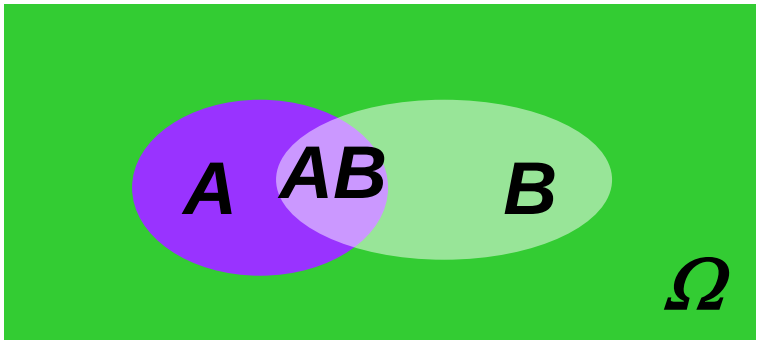
Section 2 Operations and relations of events

(**Disjoint or Mutually exclusive events**) Two events A and B are mutually exclusive, or disjoint, if $A \cap B = \Phi$. that is, if A and B have no elements in common. This means that event A and event B can't occur at the same time.

Tossing a fair coin, the frontage H on the top and the reverse side T on the top are disjoint events.

Section 2 Operations and relations of events

(**Intersection**) The intersection of two events A and B, denoted by the symbol $A \cap B = \{x | x \in A \ and \ x \in B\}$, is the event containing all elements that are common to A and B. The intersection of A and B can also be marked by A Ƀ or AB

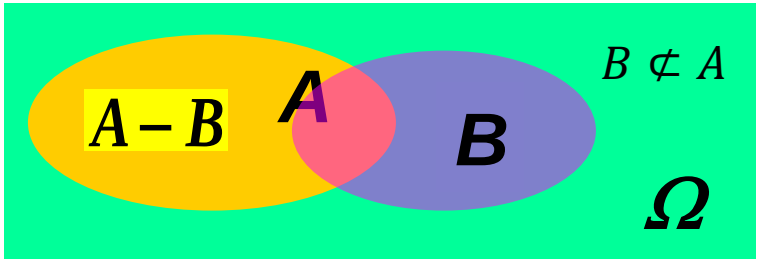


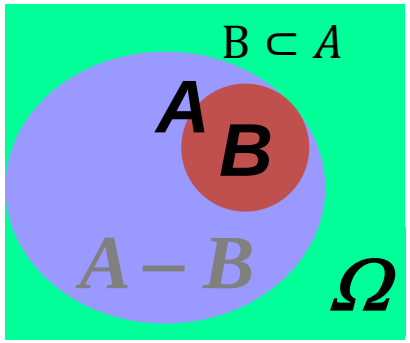
Event "a product is qualified" is the intersection of event "the length of this product is qualified" and "the radius of this product is qualified".

Note: $\bigcap_{k=1}^{n} A_k$ is the intersection of events $A_k (k = 1,2,3 \ldots)$.

Section 2 Operations and relations of events

(**Difference**) The difference of two events A and B contains all sample points that belong to A but not to B, i.e. $A - B = \{x | x \in A \ and \ x \notin B\}$.

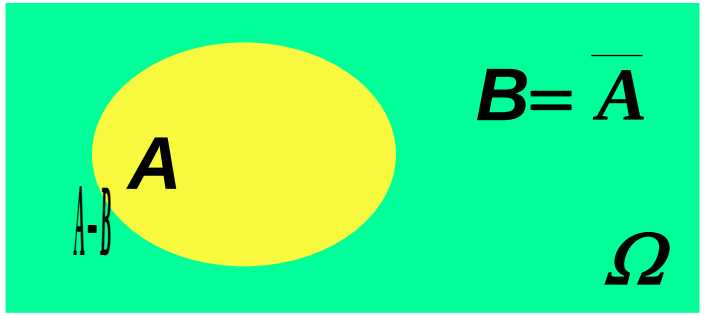




Rolling a fair die, A={1,2,3},B={2,3}, A-B=?
A-B={1};
A={1,2,3}, B={2,3,4,5,6}, A-B=?
A-B={1}

Section 2 Operations and relations of events

(**Complement**) The complement of an event A with respect to Ω is the subset of all elements of Ω that are not in A. We denote the complement of A by the symbol $\bar{A}, A', \boldsymbol{A^C}$.



Section 2 Operations and relations of events

Complement and Disjoint (Mutually exclusive) Events

disjoint

$$= \overline{A}$$

complement

$$A \cup B = \Omega \text{ and } AB = \Phi$$

disjoint $\not\rightleftarrows$ **complement**

Section 2 Operations and relations of events

Commutative law:

$$A \cup B = B \cup A,\ A \cap B = B \cap A$$

Associative law:

$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) = AB \cup AC$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) = (A \cup B)(A \cup C)$$
$$A \cap (B - C) = AB - AC$$

De Morgan's law:

$$\overline{A \cup B} = \bar{A} \cap \bar{B}; \overline{A \cap B} = \bar{A} \cup \bar{B}$$
$$\overline{\cup_{i=1}^{n} A_i} = \cap_{i=1}^{n} \overline{A_i}; \overline{\cap_{i=1}^{n} A_i} = \cup_{i=1}^{n} \overline{A_i}$$

A useful formula:

$$A - B = A - AB = A \cap \bar{B}$$

Section 2 Operations and relations of events

Example 1.2.1: assuming that A, B and C are random events, writing down the following event with A, B and C.

(1) A occurs, B and C don't.
(2) A and B occur, C doesn't.
(3) A, B and C all occur.
(4) At least one of A, B and C occurs.
(5) Neither of A, B or C occurs.
(6) No more than one event occurs.

Answer:

(1) $A\bar{B}\bar{C}$
(2) $AB\bar{C}$
(3) ABC
(4) $A \cup B \cup C$
(5) $\bar{A}\bar{B}\bar{C}$
(6) $\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + B\bar{A}\bar{C} + C\bar{B}\bar{A}$

Section 2 Operations and relations of events

Example 1.2.2: Proving $AB \cup (A - B) \cup \bar{A} = \Omega$

Answer:

$$AB \cup (A - B) \cup \bar{A}$$

$$= \mathrm{AB} \cup (A\bar{B}) \cup \bar{A}$$

$$= A(B \cup \bar{B}) \cup \bar{A}$$

$$= \mathrm{A}\Omega \cup \bar{A}$$

$$= \Omega$$